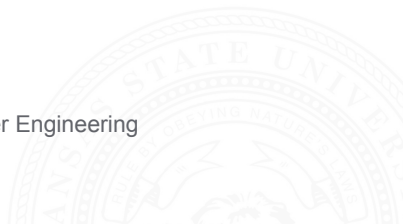


Three-Phase AC Circuits

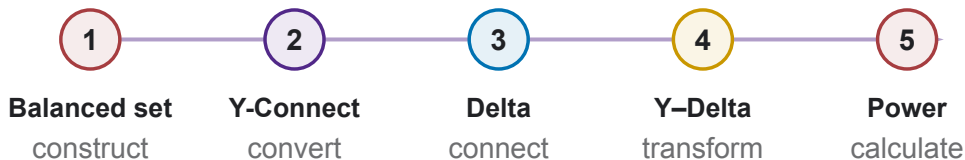
Worked examples on phase sequence, connections,
and power

Buxin She

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Lecture map



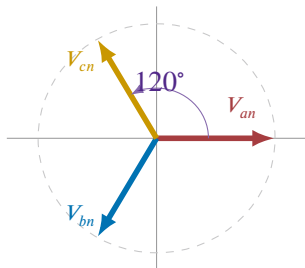
one visual summary + one complete worked example

PART 1

Balanced sets and phase sequence

A reference phasor and phase sequence determine the balanced set

Balanced set and phase sequence



equal RMS magnitude

	Positive (abc)	Negative (acb)
Phase a	$V \angle \theta$	$V \angle \theta$
Phase b	$V \angle (\theta - 120^\circ)$	$V \angle (\theta + 120^\circ)$
Phase c	$V \angle (\theta + 120^\circ)$	$V \angle (\theta - 120^\circ)$

$$|V_{an}| = |V_{bn}| = |V_{cn}|,$$

$$V_{an} + V_{bn} + V_{cn} = 0.$$

Example 1: a positive-sequence voltage set

The phase-A RMS voltage is $V_{an} = 120\angle 120^\circ$ V.

1. For a balanced positive-sequence (abc) set, find V_{bn} and V_{cn} in polar form.
2. Rotate all three phasors counterclockwise by 30° . Find V'_{an} , V'_{bn} , and V'_{cn} in polar form.
3. Convert the shifted phasors to rectangular form.
4. Verify balance using magnitudes, angular spacing, and the phasor sum.

Example 1, Q1: positive-sequence B and C voltages

Positive sequence: B lags A by 120° , and C leads A by 120° .

$$\begin{aligned}V_{bn} &= |V_{an}| \angle (\angle V_{an} - 120^\circ) \\&= 120 \angle (120^\circ - 120^\circ) \\&= \boxed{120 \angle 0^\circ \text{ V}},\end{aligned}$$

$$\begin{aligned}V_{cn} &= |V_{an}| \angle (\angle V_{an} + 120^\circ) \\&= 120 \angle (120^\circ + 120^\circ) \\&= \boxed{120 \angle 240^\circ \text{ V}}.\end{aligned}$$

Example 1, Q2: all three voltages after a $+30^\circ$ shift

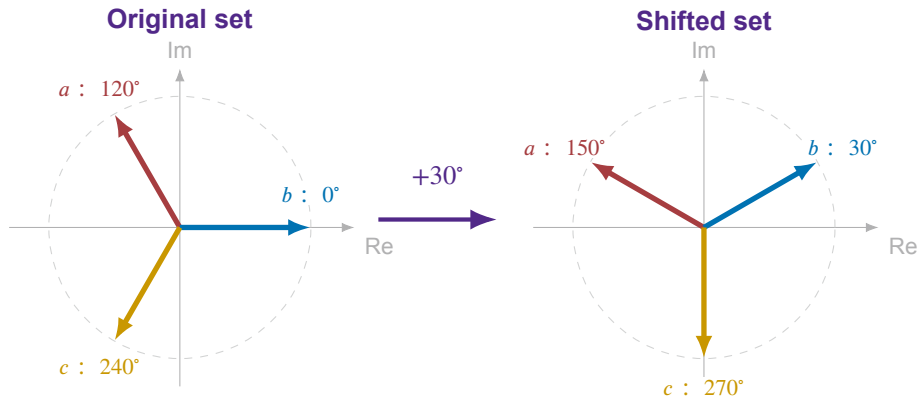
Apply $V'_{kn} = V_{kn}(1\angle 30^\circ)$ to each phase.

$$\begin{aligned} V'_{an} &= (120\angle 120^\circ)(1\angle 30^\circ) \\ &= 120\angle (120^\circ + 30^\circ) = \boxed{120\angle 150^\circ \text{ V}}, \end{aligned}$$

$$\begin{aligned} V'_{bn} &= (120\angle 0^\circ)(1\angle 30^\circ) \\ &= 120\angle (0^\circ + 30^\circ) = \boxed{120\angle 30^\circ \text{ V}}, \end{aligned}$$

$$\begin{aligned} V'_{cn} &= (120\angle 240^\circ)(1\angle 30^\circ) \\ &= 120\angle (240^\circ + 30^\circ) = \boxed{120\angle 270^\circ \text{ V}}. \end{aligned}$$

Example 1: before and after rotation



$$|V_{an}| = |V_{bn}| = |V_{cn}| = 120 \text{ V}$$

$$V'_{kn} = V_{kn}(1\angle 30^\circ), \quad k = a, b, c.$$

Example 1, Q3: shifted voltages in rectangular form

Use $V \angle \theta = V \cos \theta + jV \sin \theta$.

$$\begin{aligned} V'_{an} &= 120 \cos 150^\circ + j120 \sin 150^\circ \\ &= 120(-\sqrt{3}/2) + j120(1/2) \\ &= -60\sqrt{3} + j60 \approx \boxed{-103.923 + j60 \text{ V}}, \end{aligned}$$

$$\begin{aligned} V'_{bn} &= 120 \cos 30^\circ + j120 \sin 30^\circ \\ &= 120(\sqrt{3}/2) + j120(1/2) \\ &= 60\sqrt{3} + j60 \approx \boxed{103.923 + j60 \text{ V}}, \end{aligned}$$

$$\begin{aligned} V'_{cn} &= 120 \cos 270^\circ + j120 \sin 270^\circ \\ &= 120(0) + j120(-1) \\ &= \boxed{-j120 \text{ V}}. \end{aligned}$$

Example 1, Q4: balance verification after the shift

1. Equal RMS magnitudes

$$|V'_{an}| = |V'_{bn}| = |V'_{cn}| = 120 \text{ V.}$$

2. Equal angular spacing

$$150^\circ - 30^\circ = 120^\circ,$$

$$270^\circ - 150^\circ = 120^\circ,$$

$$(30^\circ + 360^\circ) - 270^\circ = 120^\circ.$$

3. Phasor-sum check

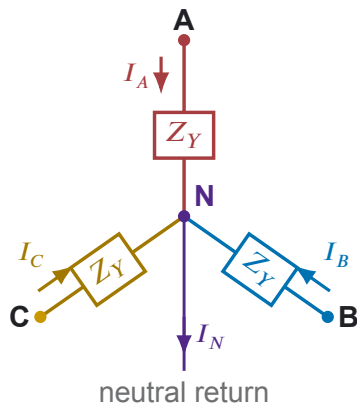
$$\begin{aligned} V'_{an} + V'_{bn} + V'_{cn} &= (-60\sqrt{3} + j60) + (60\sqrt{3} + j60) - j120 \\ &= (-60\sqrt{3} + 60\sqrt{3}) + j(60 + 60 - 120) \\ &= \boxed{0 \text{ V}}. \end{aligned}$$

PART 2

Y connection

Phase voltage across each branch; line current through each branch

Y connection: phase and line quantities



Physical quantity	Y connection
Branch voltage	V_{AN}
Line voltage	$V_{AB} = V_{AN} - V_{BN}$
Line = branch current	$I_A = V_{AN}/Z_Y$
Neutral return	$I_N = I_A + I_B + I_C$

Balanced positive sequence

$$V_{AB} = \sqrt{3} V_{AN} \angle 30^\circ$$

$$V_{LL} = \sqrt{3} V_\phi, \quad I_L = I_\phi, \quad I_N = 0.$$

$V_{LL}, V_\phi, I_L, I_\phi$ denote RMS magnitudes.

Example 2: a balanced Y-connected load

Given at the load terminals

$$V_{AB} = 13,800 \angle 0^\circ \text{ V}$$

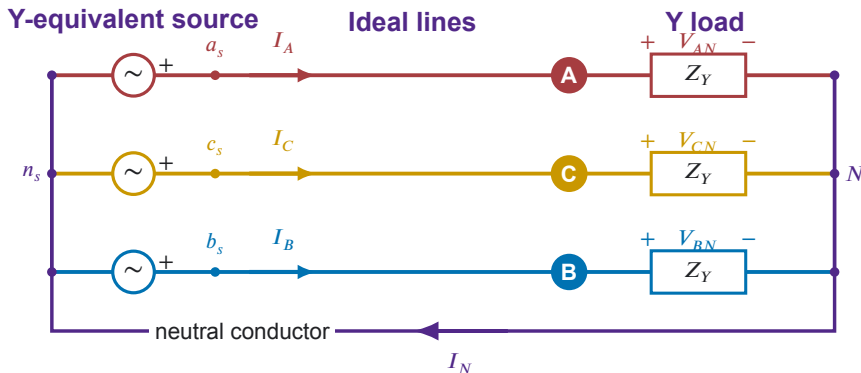
$$Z_A = Z_B = Z_C = Z_Y = 8 + j6 \, \Omega.$$

Sequence	positive (abc)
Supply lines	ideal; no voltage drop
Neutral	N connected to source n_s
Phasors	RMS values

Find, in this order

1. Convert V_{AB} to V_{AN} .
2. Complete V_{BN} and V_{CN} .
3. Calculate I_A , I_B , and I_C .
4. Calculate the neutral-return current I_N .
5. Rebuild the line voltages and verify the connection relations.

Example 2: the three-phase Y circuit



$$V_{AB} = 13.8 \angle 0^\circ \text{ kV}, \quad Z_Y = 8 + j6 \, \Omega/\text{phase}, \quad Z_\ell = 0.$$

$$I_A = I_{\text{branch},A}, \quad I_N = I_A + I_B + I_C.$$

Example 2, step 1: line voltage to phase voltage

For positive sequence, V_{AB} leads V_{AN} by 30° .

$$\begin{aligned} V_{AB} &= \sqrt{3} V_{AN} \angle 30^\circ, \\ V_{AN} &= \frac{V_{AB}}{\sqrt{3} \angle 30^\circ} \\ &= \frac{13,800}{\sqrt{3}} \angle (0^\circ - 30^\circ) \\ &= \boxed{7,967.434 \angle (-30^\circ) \text{ V}}. \end{aligned}$$

$$V_{AN} = 7,967.434 [\cos(-30^\circ) + j \sin(-30^\circ)] = 6,900 - j3,983.717 \text{ V}.$$



Example 2, step 2: complete the phase-voltage set

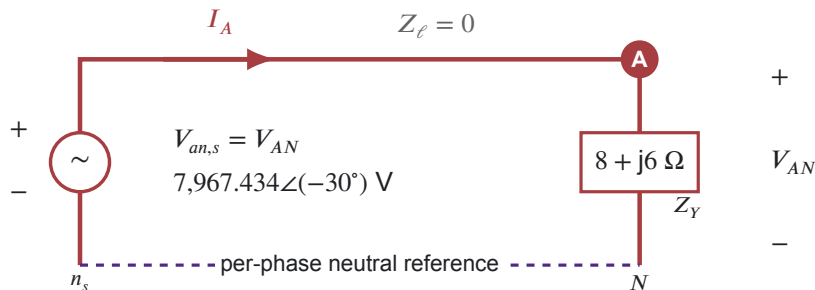
Equal magnitudes; phase B lags A by 120° , and phase C leads A by 120° .

$$\begin{aligned}V_{BN} &= 7,967.434 \angle (-30^\circ - 120^\circ) \\&= 7,967.434 [\cos(-150^\circ) + j \sin(-150^\circ)] \\&= \boxed{-6,900 - j3,983.717 \text{ V}},\end{aligned}$$

$$\begin{aligned}V_{CN} &= 7,967.434 \angle (-30^\circ + 120^\circ) \\&= 7,967.434 [\cos 90^\circ + j \sin 90^\circ] \\&= \boxed{j7,967.434 \text{ V}}.\end{aligned}$$

$$V_{AN} + V_{BN} + V_{CN} = (6,900 - 6,900) + j(-3,983.717 - 3,983.717 + 7,967.434) = 0.$$

Example 2: the phase-A equivalent circuit



$$I_A = \frac{V_{AN}}{Z_Y}, \quad \text{line current} = \text{Y-branch current.}$$

Example 2, step 3: solve the phase-A current

Convert the branch impedance

Line current = branch current

$$\begin{aligned}|Z_Y| &= \sqrt{8^2 + 6^2} = 10 \, \Omega, \\ \theta_Z &= \tan^{-1}(6/8) = 36.870^\circ, \\ Z_Y &= 10 \angle 36.870^\circ \, \Omega.\end{aligned}$$

$$\begin{aligned}I_A &= \frac{V_{AN}}{Z_Y} \\ &= \frac{7,967.434 \angle (-30^\circ)}{10 \angle 36.870^\circ} \\ &= \frac{7,967.434}{10} \angle (-30^\circ - 36.870^\circ) \\ &= \boxed{796.743 \angle (-66.870^\circ) \, \text{A}}.\end{aligned}$$

$$I_A = 796.743 [\cos(-66.870^\circ) + j \sin(-66.870^\circ)] \approx 312.977 - j732.697 \, \text{A}.$$

The current lags its own phase voltage by 36.870° , not 66.870° .

Example 2, step 4: other line currents and neutral

The three identical impedances preserve the positive sequence.

$$I_B = \frac{7,967.434 \angle (-150^\circ)}{10 \angle 36.870^\circ} = 796.743 \angle (-186.870^\circ) \text{ A},$$

$$I_C = \frac{7,967.434 \angle 90^\circ}{10 \angle 36.870^\circ} = 796.743 \angle 53.130^\circ \text{ A}.$$

$$I_B = 796.743 [\cos(-186.870^\circ) + j \sin(-186.870^\circ)] \approx -791.023 + j95.303 \text{ A},$$

$$I_C = 796.743 [\cos 53.130^\circ + j \sin 53.130^\circ] \approx 478.046 + j637.395 \text{ A}.$$

$$\begin{aligned} I_N &= I_A + I_B + I_C \\ &= I_A \left[1 + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) \right] \\ &= I_A(0 + j0) = \boxed{0 \text{ A}}. \end{aligned}$$

Example 2, step 5: rebuild and verify line voltages

$$\begin{aligned}V_{AB} &= V_{AN} - V_{BN} \\&= (6,900 - j3,983.717) - (-6,900 - j3,983.717) \\&= \boxed{13,800 + j0 = 13,800\angle 0^\circ \text{ V}},\end{aligned}$$

$$\begin{aligned}V_{BC} &= (-6,900 - j3,983.717) - j7,967.434 \\&= -6,900 - j11,951.151 = 13,800\angle(-120^\circ) \text{ V},\end{aligned}$$

$$\begin{aligned}V_{CA} &= j7,967.434 - (6,900 - j3,983.717) \\&= -6,900 + j11,951.151 = 13,800\angle 120^\circ \text{ V}.\end{aligned}$$

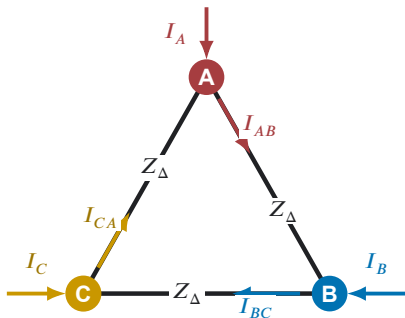


PART 3

Delta connection

Branch voltage first, then branch current and terminal KCL

Delta connection: branch quantities and line currents



Quantity	Symbol	Relation
Branch voltage	V_{AB}	$V_\phi = V_{LL}$
Branch current	I_{AB}	$I_{AB} = V_{AB}/Z_\Delta$

$$I_A = I_{AB} - I_{CA},$$

$$I_B = I_{BC} - I_{AB},$$

$$I_C = I_{CA} - I_{BC}.$$

$$I_A + I_B + I_C = 0.$$

Positive sequence (abc)

$$I_A = \sqrt{3}I_{AB}\angle(-30^\circ)$$

$$|I_A| = \sqrt{3}|I_{AB}|.$$

Example 3: balanced 13.8-kV delta load

Given and referenced

$$V_{AB} = 13.8\angle 0^\circ \text{ kV}, \quad Z_{\Delta} = 100\angle 20^\circ \Omega.$$

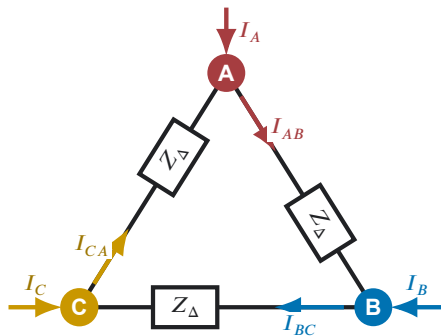
Positive sequence and negligible line impedance.

$$I_{AB} : A \rightarrow B, \quad I_{BC} : B \rightarrow C, \\ I_{CA} : C \rightarrow A.$$

Find and verify

1. Construct all three branch voltages.
2. Calculate all three branch currents.
3. Use rectangular KCL to calculate I_A .
4. Reconstruct I_B and I_C .
5. Verify the delta connection factor and the line-current sum.

Example 3: delta branches and voltage polarities



Voltage reference on each branch

Voltage	+ terminal	- terminal
V_{AB}	A	B
V_{BC}	B	C
V_{CA}	C	A

$$V_{AB} = 13.8 \angle 0^\circ \text{ kV.}$$

$$Z_{\Delta} = 100 \angle 20^\circ \Omega/\text{leg.}$$

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}}.$$

Example 3, step 1: branch-voltage set

The given phasor is $V_{AB} = 13.8\angle 0^\circ$ kV. Positive sequence rotates the remaining line voltages by -120° and $+120^\circ$.

$$V_{AB} = 13.8\angle 0^\circ \text{ kV},$$

$$V_{BC} = V_{AB}\angle(-120^\circ) = \boxed{13.8\angle(-120^\circ) \text{ kV}},$$

$$V_{CA} = V_{AB}\angle(+120^\circ) = \boxed{13.8\angle 120^\circ \text{ kV}}.$$

Each voltage is paired with the branch having the same ordered terminal labels.

Example 3, step 2: branch-current set

Convert 13.8 kV to 13,800 V before dividing by ohms.

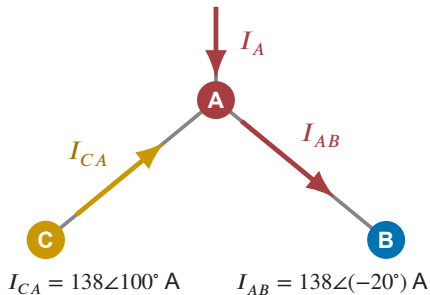
$$I_{AB} = \frac{13,800 \angle 0^\circ}{100 \angle 20^\circ} = \boxed{138 \angle (-20^\circ) \text{ A}},$$

$$I_{BC} = \frac{13,800 \angle (-120^\circ)}{100 \angle 20^\circ} = \boxed{138 \angle (-140^\circ) \text{ A}},$$

$$I_{CA} = \frac{13,800 \angle 120^\circ}{100 \angle 20^\circ} = \boxed{138 \angle 100^\circ \text{ A}}.$$

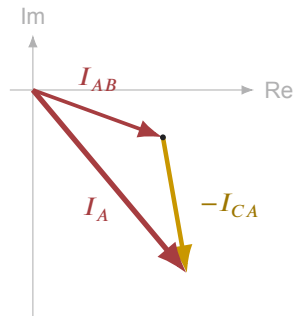
Example 3: node A and phasor subtraction

Physical current directions



$$\underbrace{I_A + I_{CA}}_{\text{enter node A}} = \underbrace{I_{AB}}_{\text{leave node A}}$$

Phasor subtraction



$$I_A = I_{AB} - I_{CA}$$

Example 3, step 3: line current from KCL

At terminal A, $I_A = I_{AB} - I_{CA}$. Convert the two branch currents:

$$I_{AB} = 138(\cos(-20^\circ) + j \sin(-20^\circ)) = 129.678 - j47.199 \text{ A},$$

$$I_{CA} = 138(\cos 100^\circ + j \sin 100^\circ) = -23.963 + j135.903 \text{ A}.$$

$$\begin{aligned} I_A &= (129.678 - j47.199) - (-23.963 + j135.903) \\ &\approx 153.641 - j183.102 \text{ A}. \end{aligned}$$

$$|I_A| \approx \sqrt{153.641^2 + 183.102^2} \approx 239.023 \text{ A},$$

$$\angle I_A \approx \tan^{-1}(-183.102/153.641) \approx -50.000^\circ,$$

$$I_A \approx \boxed{239.023 \angle (-50^\circ) \text{ A}}.$$

Example 3, step 4: complete current set and checks

$$I_A = 239.023 \angle (-50^\circ) \text{ A},$$

$$I_B = I_A \angle (-120^\circ) = 239.023 \angle (-170^\circ) \text{ A},$$

$$I_C = I_A \angle (+120^\circ) = 239.023 \angle 70^\circ \text{ A}.$$

Connection-factor check

$$\frac{|I_A|}{|I_{AB}|} = \frac{239.023}{138} \approx 1.732 \approx \sqrt{3}.$$

$$\angle I_A - \angle I_{AB} = -50^\circ - (-20^\circ) = -30^\circ.$$

Balanced-set check

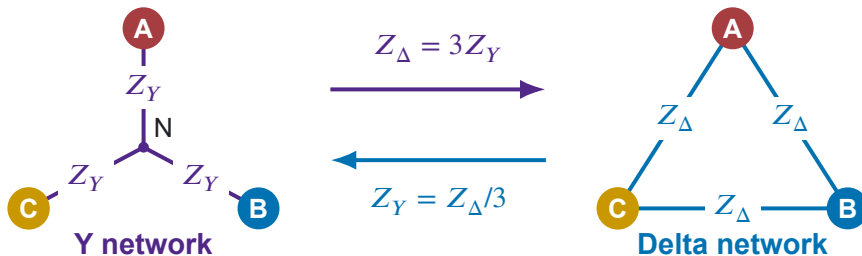
$$\begin{aligned} I_A + I_B + I_C &\approx (153.641 - 235.392 + 81.751) \\ &\quad + j(-183.102 - 41.506 + 224.608) \\ &= 0 \text{ A}. \end{aligned}$$

PART 4

Y–Delta transformation

Equivalent impedances at the same three terminals

Balanced Y–Delta transformation



Three equal impedances in each network. The Y neutral is internal.
Compare the two alternatives at the same A, B, C terminals.

Y–Delta: terminal and branch quantities

Same balanced terminal voltages	Y network	Delta network
Impedance per branch	Z_Y	$3Z_Y$
Branch voltage magnitude	$V_{LL}/\sqrt{3}$	V_{LL}
Branch current magnitude	I_L	$I_L/\sqrt{3}$
Line-current phasor	$I_A^{[Y]}$	$I_A^{[\Delta]} = I_A^{[Y]}$
Total complex power	$S_{3\phi}^{[Y]}$	$S_{3\phi}^{[\Delta]} = S_{3\phi}^{[Y]}$

$$I_A^{[\Delta]} = \frac{V_{AB} - V_{CA}}{Z_{\Delta}} = \frac{3V_{AN}}{Z_{\Delta}},$$

$$I_A^{[Y]} = \frac{V_{AN}}{Z_Y} \implies \boxed{Z_{\Delta} = 3Z_Y}.$$

Example 4: an equivalent delta for a Y load

Same load as Example 2

$$V_{AB} = 13,800\angle 0^\circ \text{ V.}$$

$$Z_Y = 8 + j6 \, \Omega \quad \text{per phase.}$$

Balanced abc, ideal lines.

Y load with floating neutral.

Questions

1. Find the equivalent Z_Δ .
2. Calculate the delta branch currents I_{AB} and I_{CA} .
3. Find $I_A^{[\Delta]}$ by KCL and compare it with $I_A^{[Y]}$.
4. Convert the delta back to Y.

Example 4, Q1: Y-to-delta impedance conversion

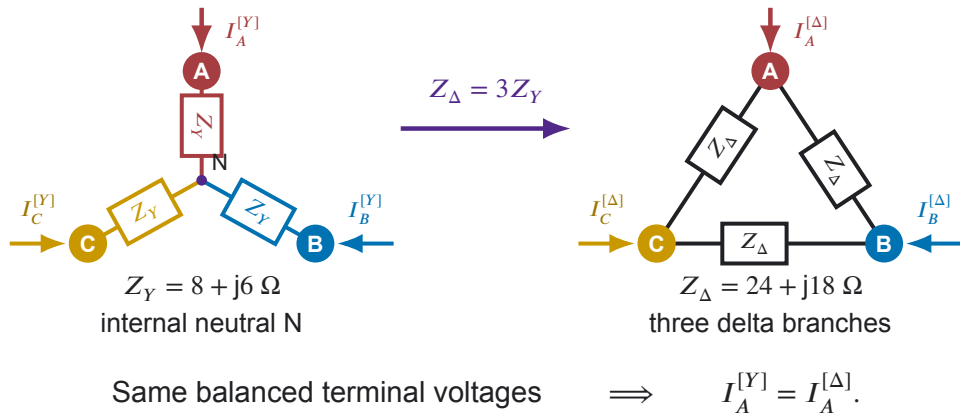
Each delta branch must have three times the impedance of one Y branch.

$$\begin{aligned}Z_{\Delta} &= 3Z_Y \\&= 3(8 + j6) \\&= \boxed{24 + j18 \, \Omega}.\end{aligned}$$

$$\begin{aligned}|Z_{\Delta}| &= \sqrt{24^2 + 18^2} = 30 \, \Omega, \\ \angle Z_{\Delta} &= \tan^{-1}(18/24) = 36.870^\circ.\end{aligned}$$

$$\boxed{Z_{\Delta} = 30 \angle 36.870^\circ \, \Omega \quad \text{per delta branch}.}$$

Example 4: the two equivalent load connections



Example 4, Q2: delta branch currents

Positive sequence gives $V_{CA} = 13,800\angle 120^\circ$ V.

$$\begin{aligned} I_{AB} &= \frac{V_{AB}}{Z_{\Delta}} = \frac{13,800\angle 0^\circ}{30\angle 36.870^\circ} \\ &= 460\angle(-36.870^\circ) \\ &= 460[\cos(-36.870^\circ) + j\sin(-36.870^\circ)] = \boxed{368.000 - j276.000 \text{ A}}, \end{aligned}$$

$$\begin{aligned} I_{CA} &= \frac{V_{CA}}{Z_{\Delta}} = \frac{13,800\angle 120^\circ}{30\angle 36.870^\circ} \\ &= 460\angle 83.130^\circ \\ &= 460[\cos 83.130^\circ + j\sin 83.130^\circ] \approx \boxed{55.023 + j456.697 \text{ A}}. \end{aligned}$$

Example 4, Q3: line-current comparison

Delta: terminal KCL

$$\begin{aligned} I_A^{[\Delta]} &= I_{AB} - I_{CA} \\ &\approx (368.000 - j276.000) \\ &\quad - (55.023 + j456.697) \\ &= \boxed{312.977 - j732.697 \text{ A}}. \end{aligned}$$

Y: phase-voltage calculation

$$\begin{aligned} V_{AN} &= \frac{13,800}{\sqrt{3}} \angle(-30^\circ) \\ &= 6,900 - j3,983.717 \text{ V}. \\ I_A^{[Y]} &= \frac{6,900 - j3,983.717}{8 + j6} \\ &= \frac{(6,900 - j3,983.717)(8 - j6)}{8^2 + 6^2} \\ &\approx \boxed{312.977 - j732.697 \text{ A}}. \end{aligned}$$

$$\boxed{I_A^{[\Delta]} = I_A^{[Y]}}, \quad |I_A| = 796.743 \text{ A}, \quad |I_{AB}| = 460.000 \text{ A}.$$

Example 4, Q4: delta-to-Y conversion

Start with the equivalent delta from Q1: $Z_{\Delta} = 24 + j18 \Omega$ per leg.

$$\begin{aligned} Z_Y^{\text{recovered}} &= \frac{Z_{\Delta}}{3} \\ &= \frac{24 + j18}{3} \\ &= \frac{24}{3} + j\frac{18}{3} \\ &= \boxed{8 + j6 \Omega}. \end{aligned}$$

Original Y	Equivalent delta	Recovered Y
$8 + j6 \Omega$	$24 + j18 \Omega$	$8 + j6 \Omega$

PART 5

Three-phase power

Use matched voltage-current pairs at each circuit boundary

Balanced three-phase power

Physical quantity	Y-connected load	Δ -connected load
Branch voltage	V_{AN}	V_{AB}
Branch current	I_A	I_{AB}
Complex power	$S_{3\phi} = 3V_{AN}I_A^*$	$S_{3\phi} = 3V_{AB}I_{AB}^*$
Magnitude relation	$V_{LL} = \sqrt{3} V_{AN} $	$I_L = \sqrt{3} I_{AB} $

$$|S_{3\phi}| = \sqrt{3}V_{LL}I_L$$

$$\phi = \angle V - \angle I$$

$$P = |S| \cos \phi, \quad Q = |S| \sin \phi$$

V and I refer to the same branch.

Source, line, and load power

Source delivery

$$S_s = 3V_{an,s}I_A^*$$

Line absorption

$$S_\ell = 3|I_A|^2 Z_\ell$$

Load absorption

$$\begin{aligned} S_L &= 3|I_A|^2 Z_Y \\ &= 3|I_{AB}|^2 Z_\Delta \end{aligned}$$

$$S_s = S_\ell + S_L,$$

$$P_s = P_\ell + P_L,$$

$$Q_s = Q_\ell + Q_L.$$

Example 5: power with line impedance

Use the source and equivalent delta load from Example 4.

From Example 4

$$V_{ab,s} = 13.8\angle 0^\circ \text{ kV},$$
$$Z_{\Delta} = 24 + j18 \text{ } \Omega/\text{leg}.$$

New condition

Add one line impedance per phase:

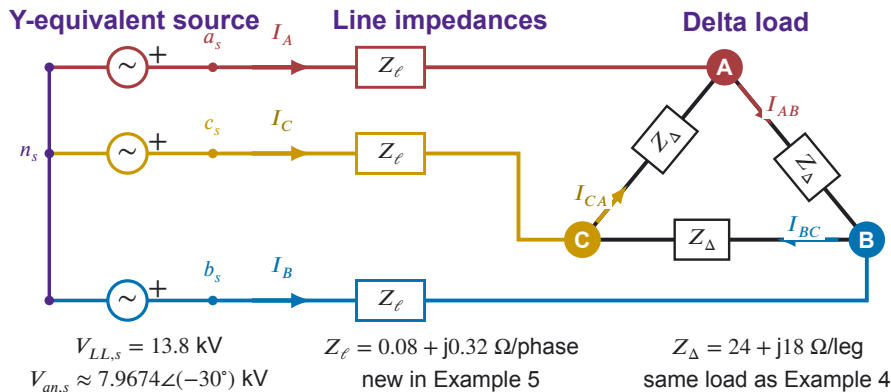
$$Z_{\ell} = 0.08 + j0.32 \text{ } \Omega.$$

Balanced abc, RMS phasors.

Questions

1. Recalculate the line current I_A .
2. Find source complex power S_s .
3. Find line absorption S_{ℓ} .
4. Find S_L by Y and delta, and $V_{LL,L}$.
5. Check $S_s = S_{\ell} + S_L$ and find $100P_{\ell}/P_s$.

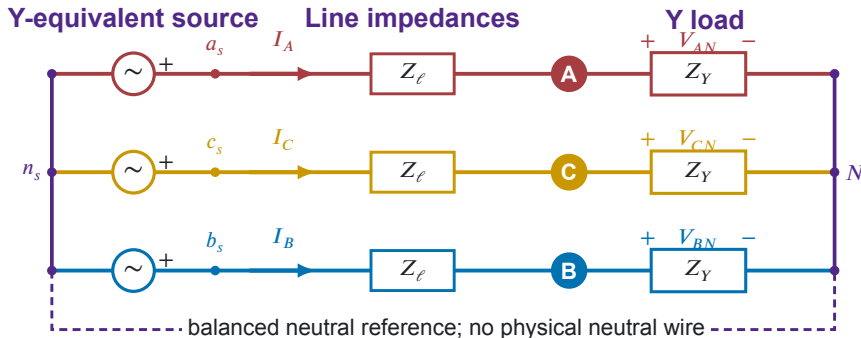
Example 5: the source–line–load circuit



$$\underbrace{S_s}_{\text{source delivery}} = \underbrace{S_\ell}_{\text{line absorption}} + \underbrace{S_L}_{\text{load absorption}}.$$

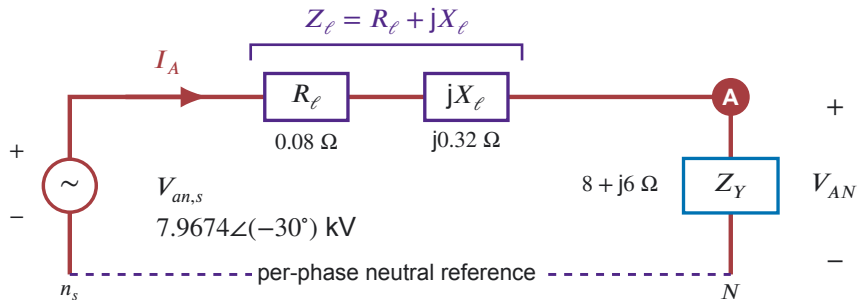
Example 5: the equivalent Y load behind the lines

$$Z_Y = \frac{Z_{\Delta}}{3} = \frac{24 + j18}{3} = 8 + j6 \, \Omega.$$



$$Z_{\ell} = 0.08 + j0.32 \, \Omega/\text{phase}, \quad Z_{\text{tot}} = Z_{\ell} + Z_Y.$$

Example 5: line impedance in the phase-A circuit



$$V_{an,s} = I_A Z_\ell + V_{AN}, \quad V_{AN} = I_A Z_Y.$$

$$I_A = \frac{V_{an,s}}{Z_\ell + Z_Y}.$$

Example 5, step 1: line current

Source and equivalent load

$$V_{an,s} = \frac{13,800}{\sqrt{3}} \angle(0^\circ - 30^\circ) \\ \approx 7,967.434 \angle(-30^\circ) \text{ V},$$

$$Z_Y = Z_\Delta/3 = 8 + j6 \, \Omega.$$

Total per-phase impedance

$$Z_{\text{tot}} = Z_\ell + Z_Y \\ = (0.08 + j0.32) + (8 + j6) \\ = 8.08 + j6.32 \, \Omega,$$

$$|Z_{\text{tot}}| = \sqrt{8.08^2 + 6.32^2} \approx 10.258109 \, \Omega, \\ \theta_Z = \tan^{-1}(6.32/8.08) \approx 38.031743^\circ.$$

$$I_A = \frac{V_{an,s}}{Z_{\text{tot}}} \approx \frac{7,967.434}{10.258109} \angle(-30^\circ - 38.031743^\circ) \\ \approx \boxed{776.696146 \angle(-68.031743^\circ) \text{ A}}.$$

Example 5, step 2: source complex power

Use $S_s = 3V_{an,s}I_A^*$, retaining unrounded intermediate values.

$$\begin{aligned} S_s &= 3V_{an,s}I_A^* \\ &= 3(7,967.434)(776.696146)\angle[-30^\circ - (-68.031743^\circ)] \\ &= 18.564825\angle 38.031743^\circ \text{ MVA} \\ &= 18.564825[\cos 38.031743^\circ + j \sin 38.031743^\circ] \text{ MVA} \\ &= \boxed{14.623 + j11.438 \text{ MVA}}. \end{aligned}$$

$$P_s = 14.623 \text{ MW}, \quad Q_s = 11.438 \text{ Mvar}.$$

Example 5, step 3: three-phase line absorption

Square the recalculated RMS line-current magnitude:

$$|I_A|^2 = (776.696146)^2 = 603,256.903 \text{ A}^2.$$

$$\begin{aligned} S_\ell &= 3|I_A|^2 Z_\ell \\ &= 3(603,256.903)(0.08 + j0.32) \\ &= 144.782 + j579.127 \text{ kVA} \\ &= \boxed{0.145 + j0.579 \text{ MVA}}. \end{aligned}$$

The three line resistances absorb $P_\ell \approx 144.782 \text{ kW}$.

Example 5, step 4: load power by two routes

Equivalent-wye route

$$Z_Y = \frac{24 + j18}{3} = 8 + j6 \, \Omega.$$

$$\begin{aligned} S_L &= 3|I_A|^2 Z_Y \\ &= 3(603,256.903)(8 + j6) \\ &= 14,478.166 + j10,858.624 \text{ kVA} \\ &= \boxed{14.478 + j10.859 \text{ MVA}}. \end{aligned}$$

Physical-delta route

$$|I_{AB}| = \frac{776.696146}{\sqrt{3}} = 448.425729 \text{ A},$$

$$\begin{aligned} |I_{AB}|^2 &= 201,085.634 \text{ A}^2, \\ S_L &= 3|I_{AB}|^2 Z_\Delta \\ &= 3(201,085.634)(24 + j18) \\ &= \boxed{14.478 + j10.859 \text{ MVA}}. \end{aligned}$$

$$V_{LL,L} = \sqrt{3}|I_A Z_Y| = \sqrt{3}(776.696146)(10) = 13,452.772 \text{ V}.$$

Example 5, step 5: check the power balance

Three-phase quantity	P (MW)	Q (Mvar)
Source delivery	14.623	11.438
Line absorption	0.145	0.579
Load absorption	14.478	10.859

$$P_{\ell} + P_L = 0.145 + 14.478 = 14.623 = P_s,$$

$$Q_{\ell} + Q_L = 0.579 + 10.859 = 11.438 = Q_s.$$

Source magnitude: kV $\times$ kA

$$|S_s| = \sqrt{3}(13.8)(0.776696) \\ = 18.565 \text{ MVA.}$$

Load magnitude: kV $\times$ kA

$$|S_L| = \sqrt{3}(13.452772)(0.776696) \\ = 18.098 \text{ MVA.}$$

$$\text{Line-loss percentage} = \frac{144,781.657}{14,622,947.330} \times 100\% = \boxed{0.990\%}.$$

Complete solution workflow

- 1 Declare RMS values, sequence, voltage endpoints, and current directions.
- 2 Apply the connection relations and the balanced Y–Delta transformation at the same terminal boundary.
- 3 Calculate branch quantities, use terminal KCL for line currents, and check equivalent-network results.
- 4 Verify KVL, KCL, connection factors, and $S_s = S_\ell + S_L$ using unrounded intermediate values.